Cell Decision Making Based on Entropic Principles

Thesis submitted in partial fulfillment of
the requirements for the degree of

Bachelor of Science (Research)

By

Aditi Ajith Pujar

Sr.No. 11-01-00-10-91-17-1-14740

Under the supervision of

Dr. Haralampos Hatzikirou
Department of Mathematics

Khalifa University

Dr. Mohit Kumar Jolly
Center for BioSystems Science and Engineering
Indian Institute of Science

June, 2021

Acknowledgement

First and foremost, I would like to thank my advisors - Dr. Mohit Kumar Jolly, who first let me join his lab and gave me the leeway to find what I really liked to tinker with, which is how I found myself working on a collaborative project with Dr. Haralampos Hatzikirou, before I joined to work under him for my Bachelor's thesis.

I am so thankful to Dr. Hatzikirou for giving me the chance to work with him, and for his geniality and generosity that are matched only by his knowledge.

The deepest of thank you's to Dr. Jolly, for truly being the advisor and mentor that we all always want but don't always deserve.

I would also like to thank Prof. Prabal Maiti, for taking me under his wing as my secondary guide.

I am so grateful to Mr. Arnab Barua, one of Dr. Hatzikirou's students, for being so approachable, patient and generous with his time while also being so ridiculously good at what he does.

I am so grateful to Ushasi Roy, part of Dr. Jolly's lab, for being so approachable, and her gentle yet firm way of getting things done.

I am truly indebted to Divyoj Singh, a fellow undergraduate, labmate and friend, for first introducing me to the CSB lab, and for having been in my corner every since.

And lastly, I would like to thank my parents, and my brother. Everything that is good, finds its meaning in them.

Abstract

In the complex world of biological phenomena, complete knowledge of all the underlying mechanisms of a process is close to impossible. Thus, phenomenological ways to draw models and meaning are very helpful. Purely data-driven approaches allow for some quantitative prediction but have restricted mechanistic interpretability. Therefore, an approach that brings a certain level of mechanistic understanding and that also allows for predictions without requiring precise biological knowledge is ideal.

*

LEUP (Least microEnvironmental Uncertainty Principle) is a variational principle, that considers cells to be Bayesian decision-makers of their phenotype, and imposes the minimisation of joint entropy (uncertainty) between the cells intrinsic state and environmental variables.

It has already shown great promise in being a generative model for different biological processes, capturing much of the observed behaviours in cell migration, [2], predicting tissue-level phenotype distributions [1] and transitions similar to those seen in well documented pathways like the EMT [3].

Notch signalling is a well studied pathway that plays a crucial role in determining cell fates across numerous species [5]. When aberrant, it plays an especially predominant role in the growth and proliferation of cancers[6]. Though initially thought to give rise to only two distinct phenotypes in neighboring cells, it was shown that a third, hybrid state of the two phenotypes could also arise. [4].

Previously, LEUP had only been applied to and studied in systems with discrete phenotypes. [3].The above findings motivated the question, how would systems governed by LEUP behave, if their phenotypes were continuous variables. This project is our attempt to answer

that question.

Our Contributions

As the Notch signalling pathway is an interaction between cells in contact, we first simulated populations of cells on different 2D lattices. After gaining a thorough understanding of the LEUP systems, we attempted to draw some parallels between the phenomenological models of LEUP and the mechanistic models of the Notch signalling pathway. Thus, we evaluated how LEUP performs as a generative model, for this important class of biological behaviour.

Division of Thesis

Chapter 1 outlines the theory underlying the LEUP, and the exact equations we simulate. Chapter 2 discusses the methods and the code developed for an efficient way to work with hexagonal lattices and their various neighborhoods. Chapter 3 summarises the results while Chapter 4 describes the ongoing work, future directions and presents some concluding remarks.
Chapter 5 is the Appendix.

Contents

1 Introduction **6**

1.1 The Cell as a Bayesian Learner 6

1.1.1 LEUP: the Least microEnvironmental Uncertainty Principle 8

1.2 An Illustrative example of LEUP: the Gaussian Case 9

1.2.1 Assumption about Form of $P(Y|X = x)$ 9

1.2.2 Simulating the Langevin's Equation 10

2 Methods **13**

2.1 Time Scales and Normalisations 13

2.2 Hexagonal Lattices and Neighborhoods 14

2.2.1 Indexing and Neighborhoods in Hexagonal Lattices 14

2.2.2 Shifting the origin to an arbitrary point 15

3 Results **17**

3.1 Overarching Results: Phase Space and Phase Transitions 18

3.2 Change in $\langle\sigma^2\rangle_L$ with time t 21

3.2.1 Phase Transition 1: in $(r = 1, \beta > 0)$ 21

3.2.2 Monotonic trends, no phase transition: 21

3.3 Phase Transitions on order parameter $P(\langle x\rangle)$:
$\beta > 0$ Systems 22

3.3.1 Varying r, fixed $\beta = 25$ 22

3.3.2 Varying β, fixed r 23

3.4 Phase Transitions on order parameter $P(\langle x\rangle)$:
$\beta < 0$ Systems 25

3.5 Spatial Patterns and Parallels to the Notch Delta Pathway 26

3.5.1 Notch Delta Jagged Signalling Pathway 26

3.5.2 Systems with Increasing Variance ($\langle\sigma^2\rangle_L$) 28

3.5.3 Systems with Monotonically Decreasing Variance ($\langle\sigma^2\rangle_L$) 28

3.5.4 $r > 1$ LEUP Populations 31

4 Conclusion **32**

4.1 Discussion 32

4.2 Ongoing Work and Future Directions 33

4.2.1 Two Ligand LEUP Model 33

4.2.2 Development of Mean Field Theory 33

4.3 Concluding Remarks 35

5 Appendix **36**

Chapter 1

Introduction

A cell, when placed in any environment, will collect information from it; this is followed by a change in the internal state of the cell in response to the information it collects. Thus, in the language of information theory, chemical signals (in the form of ligands and nutrients) are encoded and taken to the nucleus of a cell (by receptors and other such cell machinery), where they are decoded, upon which the cell takes decisions.

However, as these cascades of reactions are energetically expensive and slow processes, rather than a single step drastic change in state, the cell can be said to undergo *Bayesian learning*, in which, the priors of the cell populations are continually updated by the posteriors of previous time steps; at each time step, a little of the microenvironmental information is coded into the would-be priors.

1.1 The Cell as a Bayesian Learner

To start looking at this from an information theoretical perspective, our basic assumption is that the cell is a *Bayesian decision maker*: For a single cell, let its internal variable be denoted by x, and the external variable by y (corresponding to the state of its microenvironment). The variables x, y are catch all terms for the overall states of the system and its microenvironment respectively. They could be any parameter from concentration levels of a molecular marker to the velocities of cells. [1]

[1] In this thesis, the parameter they refer to is the phenotype of the cells the biological catch-all term for the 'type' of a cell

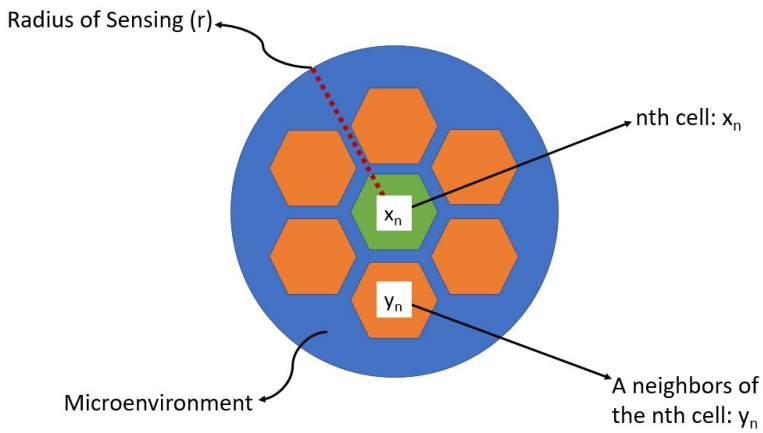


Figure 1.1: Schematic Diagram of hexagonal cells in a lattice and their microenvironment, when sensing radius $r = 1$. For the $n-th$ cell in state x_n, the y_ns it measures are the states of neighboring cells.

If a population of cells, with their states distributed in some $P_i(x(t))$ are placed in an environment of conditions $y(t)$, at some time t, we have:

$$P\left(x(t+\tau)|y(t)\right) = \frac{P(y(t)|x(t))P_i(x(t))}{P(y(t))} \tag{1.1}$$

Where, RHS:

$$\begin{aligned}
\tau &= \text{time taken by cell to take a decision} \\
P_i(x(t)) &= \text{prior probability distribution of states} \\
P(y(t)|x(t)) &= \text{probability that a cell in state x(t) will sense y(t)} \\
&\quad \text{(EMPIRICAL LIKELIHOOD)} \\
P(y(t)) &= \text{probability of the microenvironmental conditions being y(t)} \\
&\quad \text{(NORMALISATION CONSTANT)}
\end{aligned}$$

And the LHS $P\left(x(t+\tau)|y(t)\right) = P_f(x(t+\tau))$ is the final probability distribution of the population's states. Thus, the cells do *Bayesian learning* and so adapt their new phenotypes/states accordingly [1].
Now, taking the logarithm and integrating both sides:[2]

[2]The entropies we obtain are Shannon entropies, $S(X) = -\sum p(\xi)\log p(\xi)$ are denoted over random variables.

$$
\begin{aligned}
P(x_{t+\tau}) &= \frac{P(y_t|x_t)P(x_t)}{P(y_t)} \\
\log\big(P(x_{t+\tau})\big) &= \log\big(P(y_t|x_t)\big) + \log\big(P(x_t)\big) - \log\left(P(y_t)\right) \\
\implies \int \Big[\log\big(P(x_{t+\tau})\big) - \log\big(P(x_t)\big)\Big] P(x_t, y_t) dxdy &= \int \Big[\log\big(P(y_t|x_t)\big) - \log\big(P(y_t)\big)\Big] P(x_t, y_t) dxdy \\
\implies S(X_{t+\tau}) - S(X_t) &= S(Y_t|X_t) - S(Y_t) = -I(X_t, Y_t) \\
\implies \frac{S(X_{t+\tau}) - S(X_t)}{\tau} &= \frac{-I(X_t, Y_t)}{\tau}
\end{aligned}
$$

Therefore, for $\tau \ll 1$

$$
\boxed{\frac{\partial S(X|Y = y_t)}{\partial t} = \frac{\partial S(X_f)}{\partial t} = \frac{-I(X_t, Y_t)}{\tau}} \tag{1.2}
$$

where $I(X_t, Y_t)$ is the *mutual information*. As this is always a non-negative quantity, we can tell that as system reaches equilibrium,

$$
\dot{S}(X) < 0 \ \ \forall\, t
$$

or $S(X)$ is a decreasing quantity. (Here, x_t, y_t are some values/realisaions of the random variables X, Y.)

1.1.1 LEUP: the Least microEnvironmental Uncertainty Principle

Now, we assume that $\tau \ll \mathcal{T}$ where $\mathcal{T}$ is the timescale of biological processes that cause significant changes in microenvironment y (such as cell division). This is biologically valid (e.g. in humans, the cell cycle takes $\sim$ 24 hours while most molecular reactions settle to equilibrium in the time scale of $\sim$ 1 hour). This implies that $S(Y|X)$ varies very slowly.
At each time step, the cells will try to maximise their $S(X)$ (due to the maximum entropy principle), during which, due to the timescale separations, $S(Y|X)$ is taken as a constant by the cells. This imposes a constraint of $\bar{S}(Y|X)$ on the maximisation of $S(X)$. The other constraint on the system is that $p(x)$ must be normalised. Therefore, these two constraints are incorporated into the variational equation with Lagrange multipliers β and λ respectively.

At each time point, the most optimal population distributution $p(x)$ can be solved for from

the following variational equation:

$$\frac{\delta}{\delta p(x)}\left[S(X)-\beta\left(\int S(Y|X=x)p(x)dx-\bar{S}(Y|X)\right)-\lambda\left(\int p(x)dx-1\right)\right]=0$$
$$\implies \log p(x)-1-\beta S(Y|X=x)-\lambda=0$$

Therefore,

$$\boxed{p_f(x)=\frac{e^{-\beta S(Y|X=x)}}{\mathcal{Z}}} \tag{1.3}$$

The above equation is evocative of the Gibbs-Boltzmann probability disribution for a state x, where $S(Y|X=x)$ can be interpreted as the *potential* of the system in that state. (Indeed, we will use this analogy to derive a Langevin equation for the phenotypes). Here, β seems to play the role of an inverse temperature scale, but as it is the Lagrange multiplier associated with $S(Y|x)$, it can be interpreted as the *sensitivity* of the cells towards the microenvironmental sensing entropy.

1.2 An Illustrative example of LEUP: the Gaussian Case

Recalling from (1.3) that, we can treat $S(Y|X=x)$ as the **potential** of the system of cells and β is a measure of microenvironmental sensitivity.
$\implies$ the Langevin equation can therefore be written as:

$$\frac{dx}{dt}=\beta\frac{\partial S(Y|X=x)}{\partial x}+\xi \tag{1.4}$$

where, $\beta=$ measure of the microenvironmental sensitivity, $\xi=$ noise Here, $x\ \epsilon\ X$, $y\ \epsilon\ Y$ are some realisations of the random variables corresponding to internal states and microenvironmental states, respectively.

1.2.1 Assumption about Form of $P(Y|X=x)$

We make an assumption about the nature of the probability distribution, $P(y|X=x_k)$ (where x_k is some realisation): we assume that it is Normal distribution, with the mean and

variance given by the arithmetic means and variances of $y(x_k)$, i.e.

If for a cell in state x_k, N microenvironmental variables are measured are $y_1, y_2, \ldots, y_N$, with $\mu = \frac{\sum y_i}{N}$ and $\sigma^2 = \frac{\sum y_i^2}{N} - \mu^2$

$$\mathcal{P}(y) = P(y|x_k) \sim \mathcal{N}(\mu(x_k), \sigma^2(x_k))$$

The reasoning behind this is two-fold:

1. We assume that the y's vary on a much faster time scale than x_k. (though each individual y_i is the state of a neighboring cell, x_i, which must therefore vary on the same timescales as x_k), we are sampling a very large number of ys and so, the cell views correlation between each of the ys measured as going to 0
   $\implies$ the set of ys can be regarded as being *iid*, i.e. independant and identically distributed random variables.
   In this case the Central Limit Theorem tells us that their properly normalised sum is normally distributed.

2. The maximum entropy principle states that the system with the largest entropy should be chosen as the least informative default. Given (μ, σ^2), the probability distribution that satisfies these constraints and has the maximum entropy is the Normal distribution.

1.2.2 Simulating the Langevin's Equation

To calculate the RHS of the Langevin's equation for our system:

NOTATION: Assume $\mathcal{R} \epsilon \mathbb{Z}^2$ is a spatial coordinate (2D lattice with discrete positions). Then,

$$x(\mathcal{R}, t) := \text{Phenotype of cell at position } \mathcal{R} \text{ and time } t$$

*

Assumptions

1. The cells are taking decisions purely on the basis of sensing each other; there is no external forcing. i.e. for a cell in state x, the y's it measures could have have been

more complicated functions, but here,

$$y_i = \text{state of } i\text{-th neighbour} = x_i \tag{1.5}$$

A cell, x_n senses the states of N_n neighbours. Let their states be denotes as $x_1, x_2, \ldots, x_{Nn}$. The total number of cells in the neighborhood (including cell x_n) is $N := N_n + 1$.

2. As previously mentioned, we assume $\mathcal{P}(y) = P(y|X = x_n)$ is a Gaussian distribution, $\mathcal{N}(\mu, \sigma^2)$ with

$$\begin{aligned} \mu &= \frac{\sum_1^{N_n} x_i + x_n}{N} \\ \sigma^2 &= \frac{1}{N_n} \sum_1^N (x_i)^2 - \mu^2 \end{aligned}$$

Calculating microenvironmental entropy term

For a cell x_n, the probability distribution $\mathcal{P}(y) = \mathcal{N}(\mu, \sigma^2)$ has Shannon entropy

$$\begin{aligned} S(Y|X = x_n) &= \int -\mathcal{P}(y) \log\big(P(y)\big) dy \\ &= \frac{1}{2} \log K\sigma^2 \\ \implies \frac{\partial S(Y|X = x_n)}{\partial x_n} &= \frac{1}{\sigma} \frac{\partial \sigma}{\partial x_n} \end{aligned} \tag{1.6}$$

Now,

$$\begin{aligned} \text{conditional variance } \sigma^2 &= \eta \\ &= \frac{1}{N} \left(x_n^2 + \sum_{j \neq n}^{N_n} x_j^2 \right) - \frac{1}{N^2} \left(x_n + \sum_{j \neq n} x_j \right)^2 \\ &= \frac{x_n^2}{N} + \frac{1}{N} \sum_{j \neq n} x_j^2 - \frac{x_n^2}{N^2} - \frac{2x_n}{N^2} \sum_{j \neq n} x_j - \frac{1}{N^2} \left(\sum_{j \neq n} x_j \right)^2 \\ &= \frac{x_n^2}{N^2}(N-1) + \frac{1}{N} \sum_{j \neq n} x_j^2 - \frac{2x_n}{N^2} \sum_{j \neq n} x_j - \frac{1}{N^2} \left(\sum_{j \neq n} x_j \right)^2 \end{aligned}$$

Taking $N >> 1$, then $\frac{N-1}{N^2} \approx 1$. Thus,

$$\boxed{\sigma^2 = \eta = \frac{x_n^2}{N} + \frac{1}{N}\sum_{j\neq n} x_j^2 - \frac{2x_n}{N^2}\sum_{j\neq n} x_j - \frac{1}{N^2}\left(\sum_{j\neq n} x_j\right)^2} \tag{1.7}$$

So, from (1.7)

$$\frac{1}{\sigma}\frac{\partial\sigma}{\partial x_n} = \frac{1}{\sqrt{\eta}}\left(\frac{1}{2\sqrt{\eta}}\frac{\partial\eta}{\partial x_n}\right)$$

From (1.6), (1.7) and (1.8)

$$\boxed{\frac{\partial S(Y|X=x_n)}{\partial x_n} = \frac{1}{\eta}\left(\frac{x_n}{N} - \frac{1}{N^2}\sum_{j\neq n}^{N} x_j\right)} \tag{1.8}$$

Thus, we have obtained the formula used to calculate the entropic terms in our Langevin's Equation and can now proceed to simulate it.

* * *

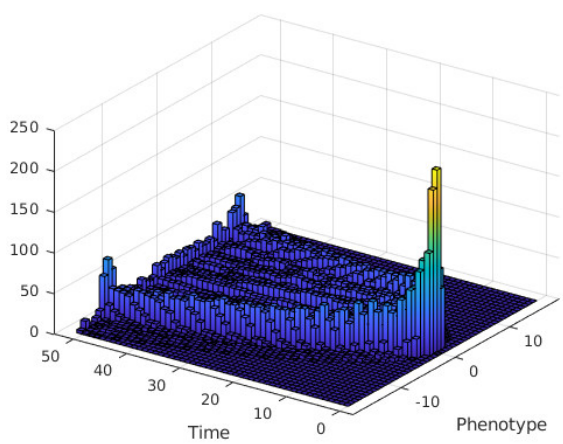
Chapter 2

Methods

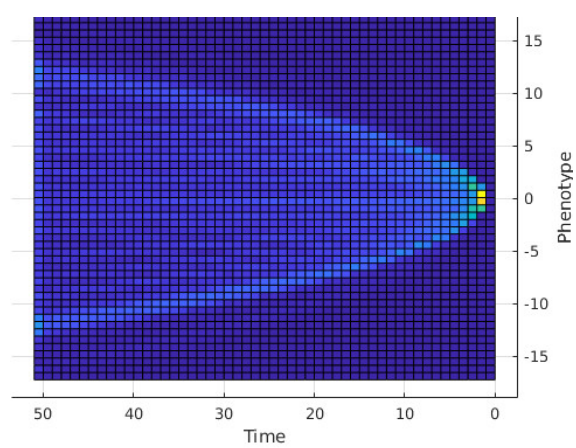
Armed with the equation 1.8, we can now simulate the Langevin equation on a lattice: for a given (r, β), at each time step, and at each point in the lattice, we find the neighbours within the radius r, implement periodic boundary conditions when required and simulate.

2.1 Time Scales and Normalisations

The first step is to characterise the time scales. Taking the same value of step size as for the Notch Delta systems, usually the total time period of simulation is defined as the point beyond which the phenotype levels (values of x) should have saturated. However, the values of LEUP phenotype do not saturate as seen in these plots:



(a) 3D Histograms, with $-Xaxis$ as time, $Y - axi$ as bins of phenotype and $Z - axis$ as frequencies.



(b) Projection of the 3D histogram onto XY plane

Figure 2.1

Though the values of LEUP do not saturate, the pattern does. Thus, beyond this, each cell will only increase the absolute magnitude of its x and no futher change takes place. Therefore, we can take the point beyond which spatial patterns stay constant as the "saturation" time(found to be $\sim TMAX = 5000$ iterations, in the videos of the heatmaps). As the variables x are anyway phenomenological, we can perform a post-facto normalisation, dividing all phenotypes by the maximum absolute value of x present in lattice. Thus, the range of phenotypes is always $\subseteq [-1, 1]$.

2.2 Hexagonal Lattices and Neighborhoods

While getting the neighborhoods and periodic boundary conditions are trivial exercises in square lattices, it is not so for hexagonal lattices. Moreover, unlike `Hexalattice` package in Python, similar toolboxes do not seem to be easily available in Matlab. So, upon studying the properties of the hexagonal lattices, codes to visualise heatmaps of hexagonal lattices as well as for more efficient way to calculate the neighborhoods for a given point (in a hexagonal lattice) could be written:

Plotting hexagonal heatmaps is fairly straightforward. Given a square NxN matrix containing 'heat' values for a hexagonal lattice, we first generate points in space where the lattice points of the hexagonal lattice would lie. A scatter plot with these points in space and the color being the matrix values, along with the Voronoi diagram (for the edges of the hexagonal unit cells) will suffice.

2.2.1 Indexing and Neighborhoods in Hexagonal Lattices

The indexing of points in hexagonal lattices can also be done in various ways. Out of the many, two are discussed.

1. **Cubic Coordinates**: A hexagonal lattice has 3 axes of symmetry. In cubic coordinates, this symmetry is exploited, each lattice point i respresented by 3 indices, (x, y, z), subject to the constraint $x + y + z = 0$. This is usually the system in which it is easiest to get the indices of neighbors (of the origin):

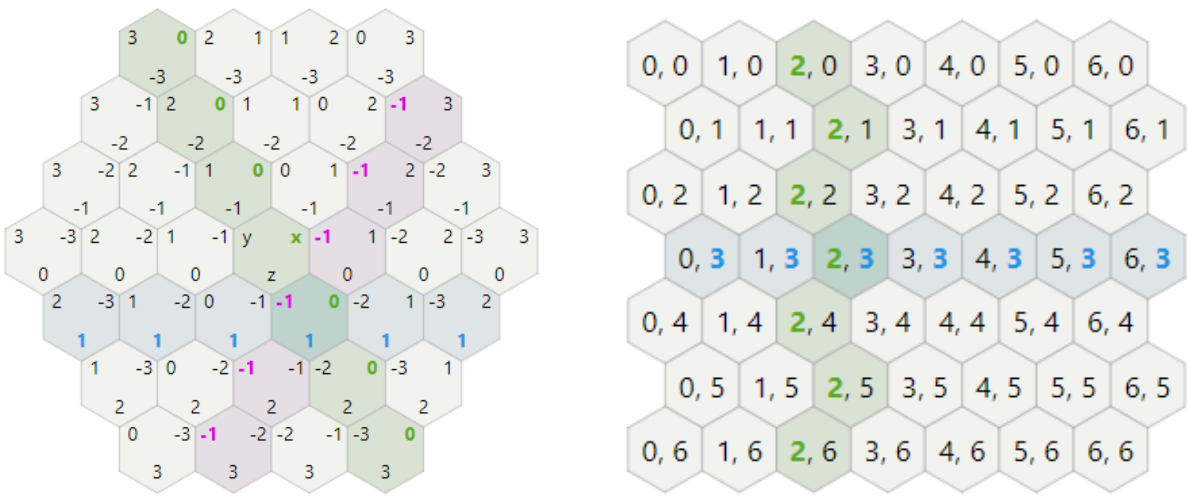
   (a) Diagonal neighbours up until a certain radius r: for $r = 1$, the 6 neighbours are the cells immediately adjacent, and for all $r > 1$, the next set of neighbours are just integral multiples of this set.

(b) Moore neighbours up until a certain radius r: all cells with their cubic indices (x, y, z) such that,

$$max[(|x|, |y|, |z|)] \leq r$$

lie within this neighborhood.

2. **Offset Coordinates**: Due to their similiarity to lattice point indices (i.e. (i, j) for a point in matrix M_{ij}), they are more easily converted to and from lattice indices and periodic boundary conditions are easy to implement as well.



(a) Cubic Coordinate Systems (b) Offset Coordinate Systems

Figure 2.2

Thus, getting the indices for different neighborhoods of the origin $(0, 0)$ is easy in the cubic coordinates. They can be transformed to offset coordinates and used to get indices of neighours for any point in the hexagonal lattice.

2.2.2 Shifting the origin to an arbitrary point

Having obtained the indices of neighbours of $(0, 0)$: first in cubic coordinates and then converted to offset coordinates/lattice coordinates, We can shift the origin of each of these indices to the point of our choice. This is again more involved than for square lattices (where simple addition suffices), and depends on the geometry of the hexagonal lattices that we have considered.

*

(These codes have been put up on GitHub: https://github.com/aditi-pujar)

This method of first getting the indices for different neighborhoods for the origin (which would be greatly aided by the symmetries of the system), and then shifting the origin of those indices to each point in the lattice as and when required, proves to be much more computationally efficient than the traditional 'brute force' method in which, the distance between the cell and a set of lattice points is calculated and all those within a given threshold will be considered as the neighbours of the cell.

Chapter 3

Results

Recalling (1.8) we are to simulate:
For a cell at point $\mathcal{R} \epsilon \mathbb{R}^2$ in space with N neighbours (in states $x_1, x_2, \ldots, x_N$), its phenotype $x_{\mathcal{R}}$ evolves with time as

$$\begin{aligned}\frac{\partial x_{\mathcal{R}}}{\partial t} &= \beta\left(\frac{\partial S(Y|X = x_{\mathcal{R}})}{\partial x_{\mathcal{R}}}\right) + \xi \\ &= \frac{\frac{x_{\mathcal{R}}}{\mathcal{N}} - \frac{1}{\mathcal{N}^2}\sum_{j\neq r}^{\mathcal{N}} x_j}{\frac{x_{\mathcal{R}}^2}{\mathcal{N}} - \frac{2x_{\mathcal{R}}}{\mathcal{N}^2}\sum_{j\neq\mathcal{R}} x_j + \frac{1}{\mathcal{N}^2}\sum_{j\neq\mathcal{R}} x_j^2 - \frac{1}{\mathcal{N}^2}\left(\sum_{j\neq\mathcal{R}} x_j\right)^2} + \xi\end{aligned}$$

(Under the assumptions that $N \gg 1$, and $\frac{N-1}{N} \approxeq 1$) From this, we can see that the 2 main parameters at our disposal are:

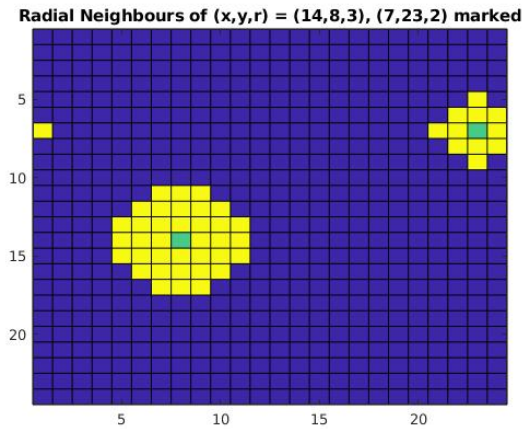
1. β: This is a measure of the cell's sensitivity to its surroundings. It could either be > 0 (< 0), interpreted as the cells maximising (minimising) their entropy. ($\beta = 0$ is a trivial case).

2. r: This is the 'radius of sensing'. i.e. we consider all the cells within a distance r as the neighbours. Thus, all their phenotypes will contribute to the RHS of 3.1 and influence the cell's decision making. Different lattices can have different neighborhoods, as seen in the previous section. We will consider the **radial neighborhoods** for *square lattices* and **Moore neighborhoods** for *hexagonal lattices* as depicted in the following figures.[1]

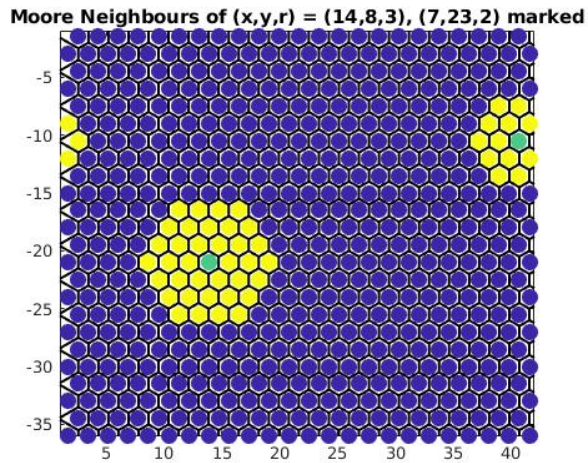
Noise is also a parameter we can vary, but it has only a trivial role to play here: incorporating

[1]The simulations were also run for other neighborhoods and the results have been included in the Appendix.

a white gaussian noise only leads to a loss of sharpness of patterns.



(a) Square Lattice



(b) Hexagonal Lattice

Figure 3.1: Neighborhoods for different r

3.1 Overarching Results: Phase Space and Phase Transitions

The figure below sums up all the results we will encounter and sets a nomenclature.

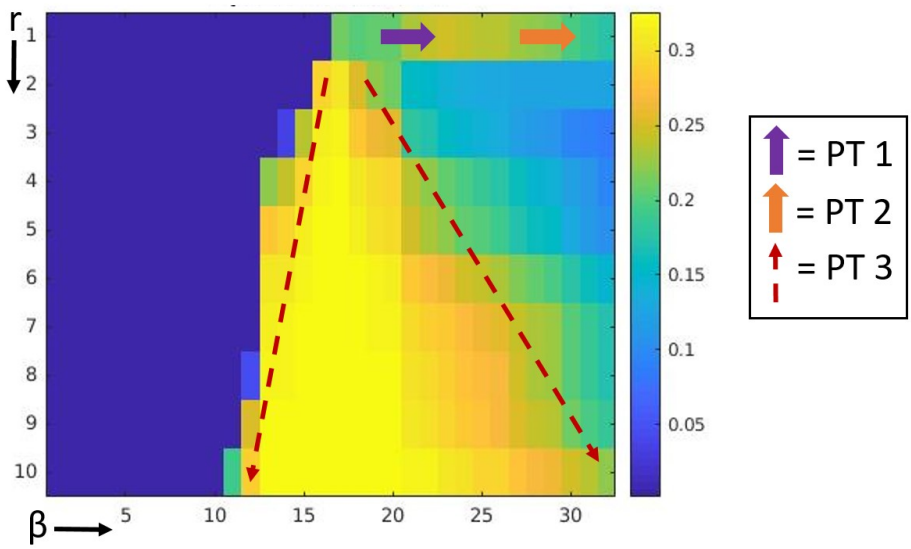
There are 2 main order parameters.

1. Order Parameter 1 $\langle\sigma^2\rangle_L$: For a cell at $\mathcal{R}$, $x_{\mathcal{R}}$,

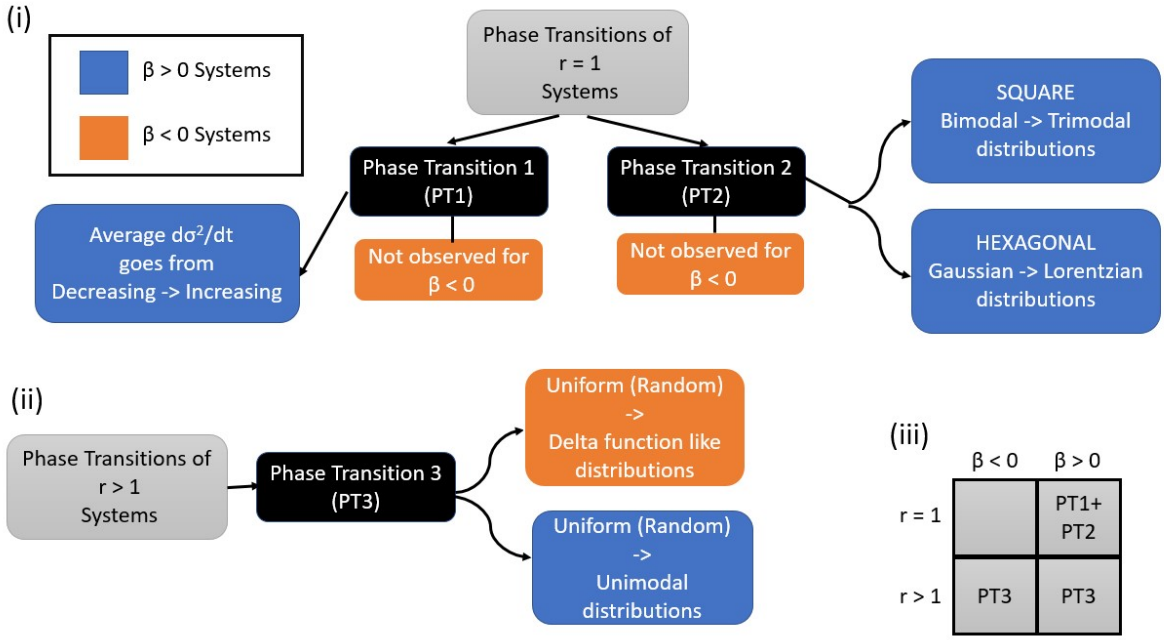
$$\begin{aligned} \sigma^2(\mathcal{R}) &= Var(y(\mathcal{R})) \text{ Arithmetic variance over y's} \\ \langle\sigma^2(\mathcal{R})\rangle &= \frac{1}{|L|}\sum_{\mathcal{R}\epsilon L}\sigma^2(\mathcal{R}) \end{aligned} \tag{3.1}$$

   where $L =$ lattice, $|L| =$ number of lattice points. This is the local variance, averaged over the entire lattice, $\langle\sigma^2\rangle_L$. This is the order parameter plotted in the first phase diagram.

2. Order Parameter 2: For $PT2$, $PT3$, the order parameter is simply the state of the cells in the lattice. Thus, the histograms are all of the probability distribution $P(\langle x\rangle_{\mathcal{R}\epsilon L})$



(a) Phase Diagram with order parameter being final average σ^2 variance over the lattice.The purple and orange arrows correspond to Phase Transitions 1, 2 respectively, while for every r, at a point along the 2 red dotted lines, Phase Transitions of type 3 are occuring. Shown for square lattice of size $25x25$, though hexagonal lattices have exact same PD.



(b) Flowchart summarising all the patterns encountered and setting up nomenclature for easy reference.**All** PTs are described as change occuring when $|\beta|$ increases.

Figure 3.2

The order parameter of interest showing all the transitions proved to be the local σ^2 averaged over all cells on the lattice $\langle\sigma^2\rangle_L$, at the end of the simulations. However, this was not immediately obvious; the first analyses were to see the population distributions of the cells,

with different indices to characterise them. The above-mentioned phase transitions were then identified. The distinction between $r = 1$ (nearest neighbour interactions) and $r > 1$ systems is very strong. For each, the βs are then varied.

1. $r = 1$ **Systems**: There are no appreciable patterns observed for $\beta < 0$ (the quantity of interest, $\frac{d\sigma^2}{dt}$ is always monotonically decreasing $\forall\ \beta < 0$)
   For $\beta > 0$:)

   (a) Phase Transition 1: When we see the $\langle\sigma^2\rangle_L$ evolving with time- for small β, $\frac{d\sigma^2}{dt}$ is initially monotonically decreasing. However, beyond a β_{PT1}, $\frac{d\sigma^2}{dt}$ starts increasing and saturates after hitting a minimum initially. This point is when the sensing ($|\beta|$) is enough to start producing patterns. This is observed in both, *square* and *hexagonal* lattices.

   (b) Phase Transition 2: This pis defined over order parameter 2, the states of the cells in the systm ($x_{\mathcal{R}},\ \ \mathcal{R}\epsilon L$)These are not observed for $\beta < 0$.
   For $\beta > 0$, as β is increased, the population distributions go from

      i. *Square Lattice:* **bimodal** (2 peaks close to extrema ($x = \pm 1$) with some amount of separation) to **trimodal** (3 equal peaks at extrema as well as hybrid $x = -1, 0, +1$). Histograms are shown in next figures.

      ii. *Hexagonal Lattice:* **Gaussian** to **Lorentzian** distributions. The tails of the distributions (of $\mathrm{P}(\langle x\rangle_{\mathcal{R}\epsilon L})$) become more pronounced.

2. $r > 1$ **Systems**: For both $\beta > 0$ and $\beta > 0$, for every value of r, Phase transitions 3 are observed. Here, populations that are seemingly uniformly randomly distributed for small β, beyond a β_{PT3} start becoming more unimodal (for $\beta > 0$) and completely localised at a single phenotype, almost a delta-function like distribution (for $\beta < 0$).

   - As r increases, $PT3$ occurs at increasing values of $|\beta_{PT3}|$.
   - These are the **only** phase transitions observed for $\beta < 0$ systems.

Now, we can look at each of the systems in detail, with a slight change in organisation: $PT1$ deals with the time evolution of σ^2's while $PT2$ and $PT3$ are characterised by population distributions. Accordingly, we first look at σ^2 evolving with time and then at $PT2$, $PT3$ in detail.

3.2 Change in $\langle\sigma^2\rangle_L$ with time t

The $\langle\sigma^2\rangle_L$ is a quantity of interest because, as seen in (3.1), we make the simplifying assumption that $P(Y|X = x)$ is a Gaussian distribution, implying that $S(Y|X = x) \sim \frac{1}{2}\log K\sigma^2$. Thus, the variance is actually an (albeit naive) measure of the microenvironment sensing entropy in the lattice. Now, the $\sigma^2(x_{\mathcal{R}})$ for a cell could be calculated over a neighborhood of any size, R, (which needn't have any relation to the r, sensing radius). However, it is shown in Appendix that plotting the variances taken over increasing R tend to converge at $R \geq 3$. Thus, in all the plots of $\langle\sigma^2\rangle_L$ time evoution, $R = 3$ is fixed.

As $|\beta|$ increases, a change in behaviour is seen only for $(r = 1, \beta > 0)$ systems while the remaining show monotonic trends:

3.2.1 Phase Transition 1: in $(r = 1, \beta > 0)$

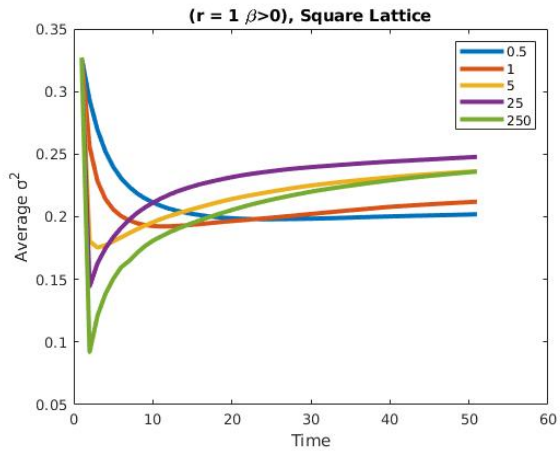
Seen in both square and hexagonal lattices:

1. For small β, σ^2 decreases and hits a saturation with time. As β increases to intermediate levels, there is a gradual increase in σ^2 after hitting a minimum. This is seen in both *square* as well as *hexagonal* lattices and is the point of $PT1$.
2. In *square lattice* further increasing β causes the $\frac{d\sigma^2}{dt} > 0$ curve to become more pronounced, and the $\sigma^2(t)$ curve, while continuous, develops a kink and is no longer differentiable. However, in *hexagonal lattice*, such drastic changes are not seen.

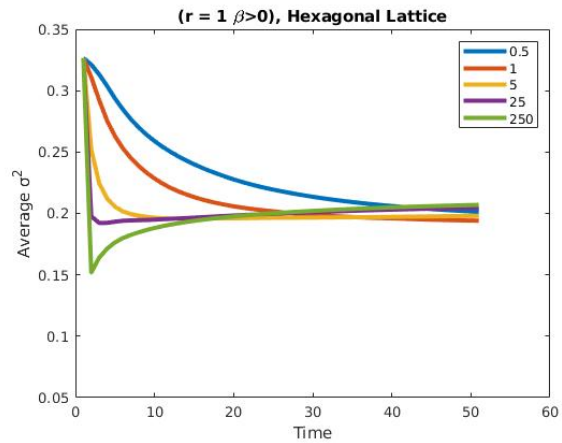
3.2.2 Monotonic trends, no phase transition:

Common to both square and hexagon lattice:

1. $(r > 1, \beta > 0)$ **Systems:** Initially $\frac{d\sigma^2}{dt} < 0$, before saturating. Increasing β causes saturation to hit faster and for lower values of σ^2.
2. $(\beta < 0)$ **Systems:** For all values of r, $\frac{d\sigma^2}{dt} < 0$ before saturation at $\sigma^2 = 0$. Higher the $|\beta|$, faster the saturation.

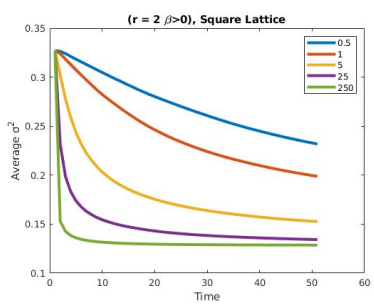


(a) $(r = 1, \beta > 0), Square$

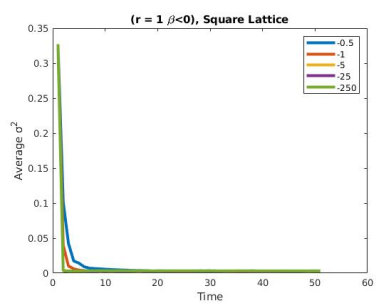


(b) $(r = 1, \beta > 0), Hexagonal$

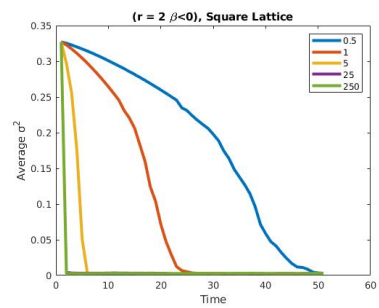
Figure 3.3: $PT1$ seen in σ^2 v/s t plots.



(a) $(r = 2, \beta > 0)$



(b) $(r = 1, \beta < 0)$



(c) $(r = 1, \beta < 0)$

Figure 3.4: Monotonic trends in $\sigma^2(t)$

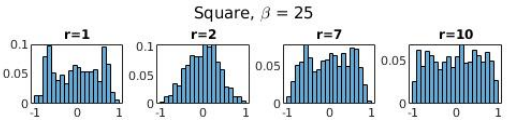
*

The above has shown us how different the $\beta > 0$ and $\beta < 0$ regimes are. Thus, while understanding patterns in population distributions, they are analysed separately.

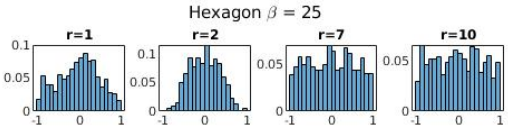
3.3 Phase Transitions on order parameter $P(\langle x \rangle)$: $\beta > 0$ Systems

3.3.1 Varying r, fixed $\beta = 25$

Initially, for a fixed sensitivity β, across multiple r:



(a) Square Lattice



(b) Hexagonal Lattice

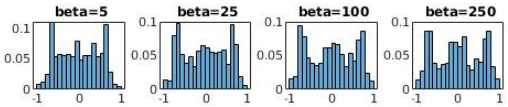
Figure 3.5: For $\beta = 25$, above histograms obtained (as r increased from left to right)

We can see many different population distributions. The difference between $r = 1$ and $r > 1$ systems is also evident - especially in case of square lattice, where

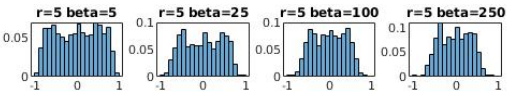
1. For $r = 1$, we have bimodal populations peaked closer to the 2 extrema (± 1)
2. At $r = 2$, the populations are centred around the "hybrid" middle state of $x = 0$

This merits a closer look at each of the systems, and understanding the role β plays.

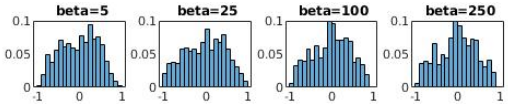
3.3.2 Varying β, fixed r



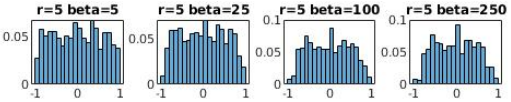
(a) Square lattice, $r = 1$



(b) Square lattice, $r = 5$



(c) Hexagonal lattice, $r = 1$



(d) Hexagonal lattice, $r = 5$

Figure 3.6: Histograms for different lattices for (LHS): $r = 1$ (RHS): $r > 1$ systems.
(β is increased from left $\rightarrow$ right)

We can see $PT2$ occuring in LHS figures and $PT3$ in the RHS figures respectively.

Phase Transition 2

1. Square Lattice: Starting with bimodal distributions with peaks nearing the extrema.[2], as β is increased, population at $x = 0$, the middle/'hybrid' phenotype, grows to form a peak. For $\beta > 100$ onwards, the height of the middle peak is $\sim$ equal to the peaks at the extrema, giving us a *trimodal* population distribution.

[2]For very small sensitivities like $\beta < 1$, the populations are uniformly random. Only beyond a critical value of β, is bimodality seen. This is indicative of Phase Transition 1

2. Hexagonal Lattice: Starting with a Gaussian distribution already centred around $x = 0$, increasing β causes the tails of the distribution to grow, i.e. the frequency of phenotypes intermediate to the extrema $x = \pm 1$ seem to decrease (but those *at* extrema may increase). This leads to Lorentzian-looking distributions for $\beta > 100$. Overall, as sensitivity is increased, the hybrid phenotype $x = 0$ seems to grow more stable/ more frequent.

Summing up,

- For low βs (beyond the sensing threshold of $PT1$), for a square lattice, the extrema phenotypes are more frequent while for hexaogonal lattices, the hybrid phenotypes are more prominently seen.
- Increasing sensitivity for square lattices cause the $x = 0$ phenotypes to grow more stable and hence more frequent, ultimately leading to populations going from bimodal $\rightarrow$ trimodal distributions.
- Similiarly for hexagonal lattices, increasing sensitivity causes phenotypes to be more centred at $x = 0$, leading to Gaussian-like populations becoming $\rightarrow$ Lorentzian-like.

While population distributions might differ, in both lattice geometries, the overarching trend is:

$$\boxed{\text{Increasing } \beta \implies \text{ Increased stability of x} = 0}$$

Phase Transition 3

We see that in both square and hexagonal lattices, $\uparrow \beta \implies$ increasing unimodality on the population distributions. This trend can be explained thus - with increasing radii of interaction,

1. Not only do the number of neighbours increase
2. As we are considering a large number of cells, over a large area: their sums will start to cancel out.

Thus, we expect that the overall sensing term of $S(Y|X = x)$ as given by 3.1 will decrease with increasing r. For the cell to still take an informed decision, a greater sensitivity (the β factor that multiplies the sensing factor), is required. Therefore, there is no sensing for lower values of *beta*, but with increasing β, the cells take informed decisions. On a population

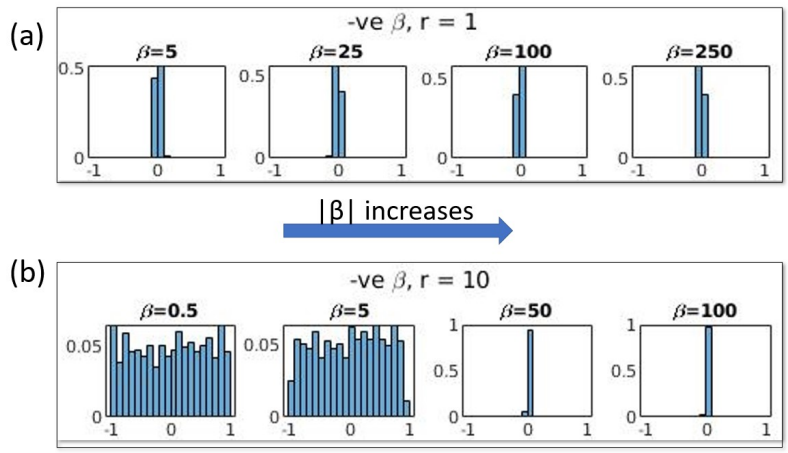


Figure 3.7: Square Lattice -ve βs for (a) $r = 1$ (b) $r = 10$
$|\beta|$ increased from left to right.

level, this appears as a transition from

$$\boxed{\text{Uniformly Random Distributions} \rightarrow \text{Unimodal populations.}}$$

This phase transition seems to be of higher order than 1, i.e. the changes are gradual, not discontinuous.

3.4 Phase Transitions on order parameter $P(\langle x \rangle)$: $\beta < 0$ Systems

In $r = 1$ systems for $\beta < 0$, almost all the cells go to the same hybrid phenotype of $x = 0$ except for a few outliers. The corresponding population distributions are therefore completely localised at $x = 0$ and have been referred to as *Delta function-like* distributions.

Performing a similar analysis as for $\beta > 0$, we can see that the dependance on the value of β is thus: We can see the same behaviours as outlined by Phase Transition 3. For large r, there is not enough sensing at low β values. Higher values of r have correspondingly higher values of β_{PT3} beyond which cells take informed decisions. For $\beta < 0$, this means:

$$\begin{aligned} \text{Not enough sensing} &\rightarrow \text{Informed Decision Making} \\ \text{Uniformly Randomly distributed populations} &\rightarrow \text{Delta function-like population distributions} \end{aligned} \tag{3.2}$$

From both, the histograms in Fig(3.7) and the phase diagram in Fig(3.2a):
PT3 for $\beta < 0$ seems more abrupt/ discontinuous[3] when compared to PT3 for +ve βs.

*

Thus, we have seen all the phase trasitions in detail. We will now look at the spatial distributions of these populations and draw parallels to the well known biological pathway, *Notch Delta Signalling.*

3.5 Spatial Patterns and Parallels to the Notch Delta Pathway

3.5.1 Notch Delta Jagged Signalling Pathway

This is the theoretical framework proposed by Jolly et al in [5], which, by virtue of incorporation of the Jagged ligand into the dynamical system of Notch and Delta, gives rise not only the two well-known distinct, complementary phenotypes but also the existence of a hybrid of these two.

1. In general, Notch-Delta systems (sans Jagged) show the phenomenon of *lateral inhibhition*, in which case 2 neighbouring cells are driven to the opposite fates.
2. Correspondingly, Notch-Jagged systems (sans Delta) exhibhit *lateral induction*, in which case all the surrounding cells are driven to the same phenotype.

The mechanism machinery consists of the Notch transmembrane receptor protein (N) and its ligands Delta(D) and Jagged(J).

- When the receptor-ligand complex is formed by N and D or J of the same cell (*cis* interactions), the complex merely undergoes degradation.
- The binding of cell's N to the ligands of another cell *trans interactions* triggers the release of Notch-Intra Cellular Domain, NICD (denoted by I), which was previously attached to the N, into the cytoplasm of the cell (the one carrying the N receptors). I then affects the levels of expression of the molecules.

[3]The color gradation is sharper

The set of ODEs are:

$$\frac{dN}{dt} = N_0\mathcal{H}^+_{\mathcal{N}}(I) - k_cN(D+J) - k_tN(D_{ext}+J_{ext}) - \gamma N \tag{3.3}$$

$$\frac{dD}{dt} = D_0\mathcal{H}^-_{\mathcal{D}}(I) - k_cND - k_tN_{ext}D - \gamma D \tag{3.4}$$

$$\frac{dJ}{dt} = J_0\mathcal{H}^+_{\mathcal{J}}(I) - k_cNJ - k_tN_{ext}J - \gamma J \tag{3.5}$$

$$\frac{dI}{dt} = k_tN(D_{ext}+J_{ext}) - \gamma_I I \tag{3.6}$$

Where,

$$(N_0, D_0, J_0) = \text{some set of basal concentrations, can be varied as parameters}$$

$$(k_c, k_t) = \text{sensitivity parameters}$$

And $\mathcal{H}$ is the shifted Hill function:

$$\mathcal{H}(x, x_0, \lambda_x, n_x) = \frac{1}{1+(\frac{x}{x_0})^{n_x}} + \lambda_x\frac{(\frac{x}{x_0})^{n_x}}{1+(\frac{x}{x_0})^{n_x}} \tag{3.7}$$

If $\lambda_x > 1 \implies H = H^+(x)$, (positively shifted, $H^+(x)$ increases with increasing x). Similiarly,
$\lambda_x < 1 \implies H = H^-(x)$, (negatively shifted Hill function.)

NOTE: Here, I_0 is a fixed value for the Hill functions and each of N,D and J have their own values of (n, λ)

(Thus, the behaviour of this system was analysed for different parameter regimes by my labmate, Mr. Divyoj Singh and some parallels to the LEUP systems were observed)

*

With many varying behaviours for different r, β, as well as a plethora of parameters for the Notch-Delta system, a *unifying lens* to classify systems comes from section (3.2):
We compare those LEUP and Notch-Delta systems with the same trends in σ^2 time evolution.

NOTE:The heatmaps for Notch Delta system are plotted using normalised values of NICD (I) as marker (dividing by maximum of absolute values of the order parameter).

3.5.2 Systems with Increasing Variance ($\langle\sigma^2\rangle_L$)

These systems have $\frac{d\sigma^2}{dt} > 0, \;\; t\epsilon\mathcal{T}$, for some interval $\mathcal{T}$. It is seen that:

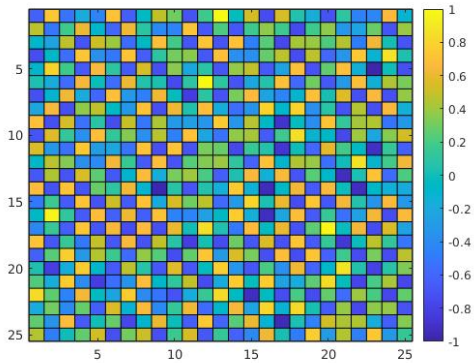
1. The square lattice cases compare quite well. A distinct chessboard like patterning (called "salt and pepper" patterning) observed in patches in both; although there is significantly higher order in the ND system. While the forms of $\sigma^2(t)$ differ, both do increase with time.

2. The parallels between hexagonal systems fall short. The distinct 'hexagon-antihexagon' patterning seen in the ND system could not be replicated by LEUP. This should be expected because even on population levels, LEUP hexagonal lattices always gave rise to unimodal populations while the above ND pattern would require bimodality. Their $\sigma^2(t)$ plots differ too, though both are increasing with time (LEUP system does so after hitting a minima first).

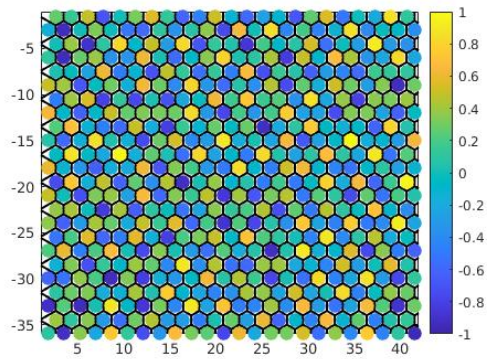
3.5.3 Systems with Monotonically Decreasing Variance ($\langle\sigma^2\rangle_L$)

Drawing parallels between the LEUP($r = 1, \beta < 0$) and ND($D_0 \approx 1400, J_0 = 1300$) regimes:[4] Here, the parallels hold up very well. In each of the systems, all the cells go to the same phenotype (save for a few outliers in the LEUP systems) and have monotonically increasing variances with time. *Parameter regime characterisation and Notch-Delta simuations have been done by Mr. Divyoj Singh, a student at Prof. Jolly's lab.*

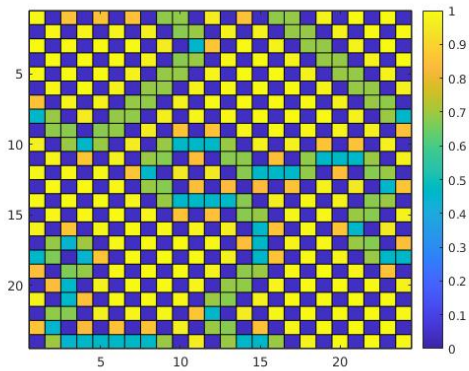
[4]Though all LEUP($r > 1, \beta > 0$) systems also have monotonically decreasing variance, these systems are considered separately because there are no analogues for $r > 1$ systems in ND signalling.



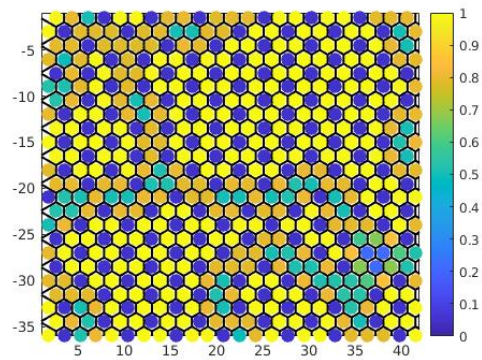
(a) $LEUP(r = 1, \beta = 25)$, Square Lattice



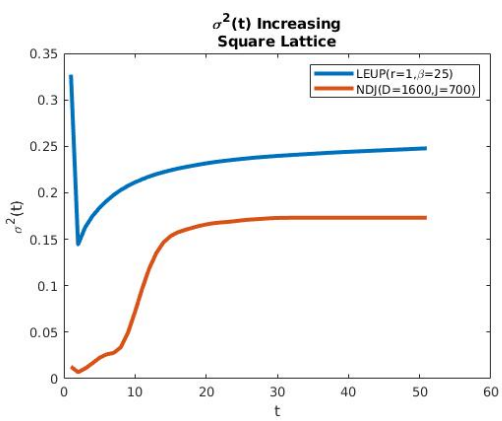
(b) $LEUP(r = 1, \beta = 25)$, Hexagonal Lattice



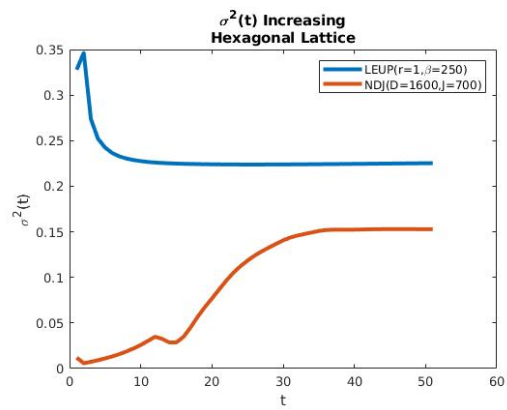
(c) $ND(D_0 = 1600, J_0 = 700)$, Square Lattice



(d) $ND(D_0 = 1600, J_0 = 700)$, Hexagon Lattice

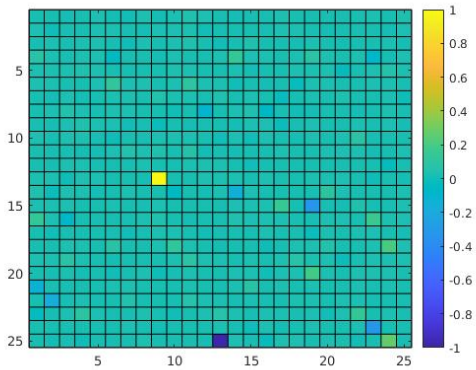


(e) σ^2 v/s t plots for both systems (Square Lattice)

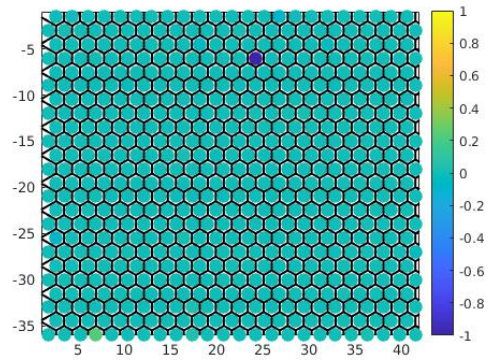


(f) σ^2 v/s t plots for both systems (Hexagonal Lattice)

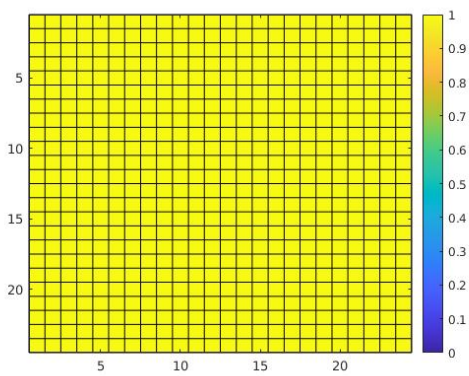
Figure 3.8: $\frac{d\sigma^2}{dt} > 0$ Systems



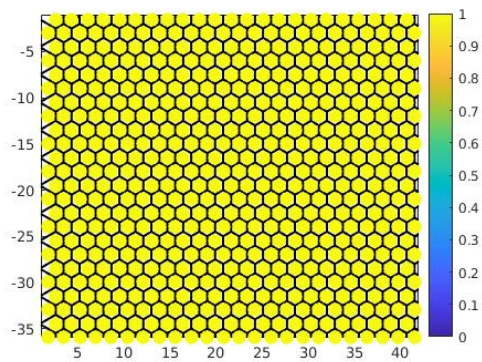
(a) $LEUP(r = 1, \beta = -25)$, Square Lattice



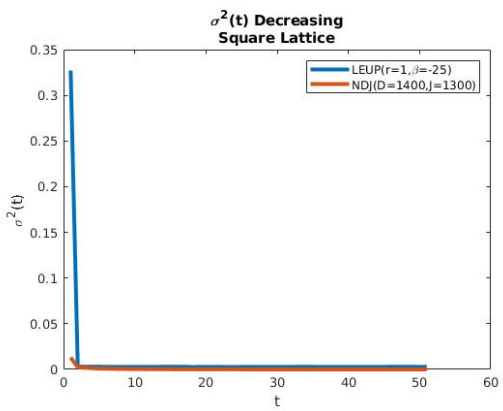
(b) $LEUP(r = 1, \beta = -25)$, Hexagonal Lattice



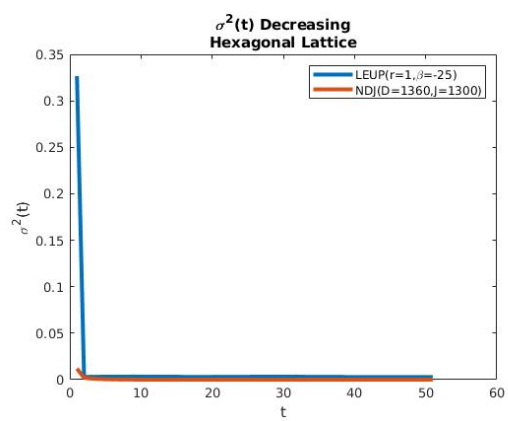
(c) $ND(D_0 = 1400, J_0 = 1300)$, Square Lattice



(d) $ND(D_0 = 1360, J_0 = 1300)$, Hexagon Lattice



(e) σ^2 v/s t plots for both systems (Square Lattice)



(f) σ^2 v/s t plots for both systems (Hexagonal Lattice)

Figure 3.9: $\frac{d\sigma^2}{dt} < 0$ Systems

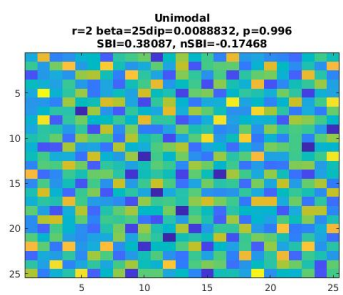
Thus, it is validating to see that a completely agnostic model such as the LEUP is able to capture some if not all, of the patterns given by the biological models of the day.

While some patterns are not captured by LEUP (e.g. hexagon-antihexagon patterning), it has also given rise to another class of patterns with no parallel in Notch Delta systems.

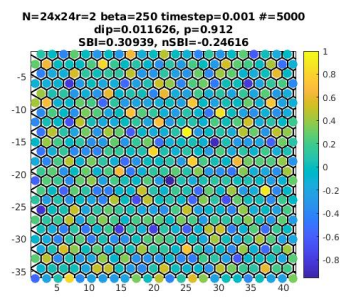
3.5.4 $r > 1$ LEUP Populations

Notch Delta signalling occurs predominantly through receptors and ligands attached to the surface of cells, making it a strictly nearest-neighbours interaction, i.e. there is no equivalent in the Notch Delta systems for an $r > 1$.

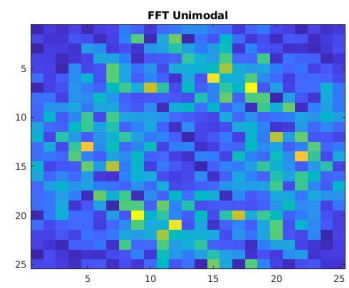
Spatial patterning: cells of extreme phenotype always seem surrounded by cells of hybrid ($x = 0$), very rarely are cells of $x = \pm 1$ immediate neighbours. This spatial patterninng is captured in their spatial FFT too, which seems to have an almost ring-like structure.



(a) Square ($r = 2, \beta = 25$)



(b) Hexagonal($r = 2, \beta = 25$)



(c) Spatial Fourier Transform of ($r = 1, \beta = 25$) Square lattice system

Figure 3.10: ($r > 1, \beta > 0$) Systems

Other biological pathways such as quorum sensing in bacteria are such that the $r > 1$ systems may have biological meaning. Thus, possible analogues with those systems can be studied futher.

* * *

Chapter 4

Conclusion

4.1 Discussion

We have built a phenomenological model of cell decision making based on the Least microEnvironemnental Entropy Principle. This was done for populations of cells on both square and hexagonal 2D lattices (for which we developed a cohesive framework in `Matlab`.

Three classes of phase transitions emerged, in different parameter regimes, each providing a different lens of insight into the nature of the LEUP governed systems.

Phase Transition 1 underscored that cells need to have some minimum sensitivity, (β) to their surroundings in order to take informed decisions. It also established that the average local variance in the populations, could be a simple, yet, very effective order parameter to not only understand these systems, but also as a means to draw parallels with others.

Phase Transition 2, in which populations with $\beta > 0$ went from having bimodal/Gaussian population distributions to trimodal/Lorentzian distributions showed that, with increasing β, the hybrid states grew in frequency; increased sensitivity correspondsed to a greater tendency to remain in more plastic states.

With Phase Transition 3, we saw that with increasing sizes of neighborhoods (increasing r), the microenvironmental signals would cancel out amongst themselves and reduce in strength, necessitating a higher sensitivity (or β value) in order to still take informed decisions.

Inspite of having no similiarities to the dynamical structure of the Notch signalling pathways, the LEUP models also displayed two of the most distinctive spatial patterns seen in the Notch Delta systems. Patterns that were not seen in Notch Delta signalling, like the unimodal

populations for $r > 0$ systems also emerged. These may have parallels in other biological pathways, underlining the wealth of behaviours captured by the LEUP models, despite their simplicity. These similiarities between the phenomenological and biological/mechanistic models have motivated a closer look, as outlined in the following section.

4.2 Ongoing Work and Future Directions

4.2.1 Two Ligand LEUP Model

The work discussed so far, has been for the simplest LEUP model possible - a population of cells, with their states characterised by a single variable x interact and take decisions that minimise their micro-environmental sensing entropy $S(Y|X = x)$. While this approach served as a generating mechanism for some of the most prominent patterns given by biological models (Notch Delta Signalling), using information from the Notch-Delta mechanism structure, we can construct a more informed model.

Recalling the Notch Delta systems described in Chapter(3): we know Notch is a receptor protein that can undergo binding to 2 different ligands, Jagged and Delta. Thus, each cell is characterised by concentrations of *two* ligands on its surface. This served as inspiration for the **2 Ligand LEUP Model**, where there are 2 random variables characterising a population of cells, i.e.

In the language of LEUP, the k_{th} cell, whose state was previously characterised by just a single variable x_k in the naive 1-ligand model, is now characterised by 2 random variables, $x_{(k,1)}$,$x_{(k,2)}$. The model is then built trying to minimise the microenvironment sensing entropy $S(Y_1, Y_2|X_1 = X_{(k,1)}, X_2 = X_{(k,2)})$.

4.2.2 Development of Mean Field Theory

Recalling (1.8), (1.7) and the associated equations, from (1.2.2) of the Introduction, we have:

$$\frac{\partial S(Y|X = x_n)}{\partial x_n} = \frac{1}{\eta}\left(\frac{x_n}{N} - \frac{1}{N^2}\sum_{j\neq n}^{N} x_j\right) \tag{4.1}$$

where,

$$\eta = \frac{x_n^2}{N} + \frac{1}{N}\sum_{j\neq n} x_j^2 - \frac{2x_n}{N^2}\sum_{j\neq n} x_j - \frac{1}{N^2}\left(\sum_{j\neq n} x_j\right)^2 \tag{4.2}$$

Recalling that N_n were the number of neighbours sensed by a cell, x_n and $N = N_n + 1$ were then the total number of cells in that neighborhood:

When $N \gg 1$ and $\frac{N-1}{N^2} \approxeq \frac{1}{N}$, we also have $N \approxeq N_n$. Substituting in above equations gives:

(4.1) implies

$$\begin{aligned}
\frac{\partial S(Y|X = x_n)}{\partial x_n} &= \frac{1}{\eta}\left(\frac{x_n}{N_n} - \frac{1}{N_n^2}\sum_{j\neq n}^{N_n} x_j\right) \\
&= \frac{1}{\eta}\left(\frac{x_n}{N_n} - \frac{1}{N_n}\frac{\sum_{j\neq n}^{N_n} x_j}{N_n}\right) \\
&= \frac{1}{\eta}\left(\frac{x_n}{N_n} - \frac{1}{N_n}(\mu)\right) \\
&= \frac{1}{\eta}\left(\frac{1}{N_n}(x_n - \mu)\right)
\end{aligned}$$

where,

$$\mu = \frac{1}{N_n}\sum_{1}^{N_n} x_j$$

And (4.2) implies

$$\begin{aligned}
\eta &= \frac{x_n^2}{N_n} + \frac{1}{N_n}\sum_{j\neq n}^{N_n} x_j^2 - \frac{2x_n}{N_n}\frac{\sum_{j\neq n} x_j}{N_n} - \frac{1}{N_n^2}\left(\sum_{j\neq n} x_j\right)^2 \\
&= \frac{x_n^2}{N_n} + \frac{1}{N_n}\sum_{j\neq n}^{N_n} x_j^2 - \frac{2x_n}{N_n}\mu - \mu^2
\end{aligned}$$

In limits where mean field theory is applicable,$< x^2 >= (< x >)^2$, thus the second and fourth terms cancel out, giving

$$\dot{x_n} = \frac{\partial S(Y|X = x_n)}{\partial x_n} = \frac{x_n - \mu}{x_n(x_n - 2\mu)} \tag{4.3}$$

From the above simplified Langevin equation, we obtain the Fokker Planck equation of the population distributions. The potentials of the system can be deduced from steady state solutions of the FKE and further analysed. *Both these models were built by Mr. Arnab Barua, a senior student at Prof. Hatzikirou's lab and my primary role has been in simulating them.*

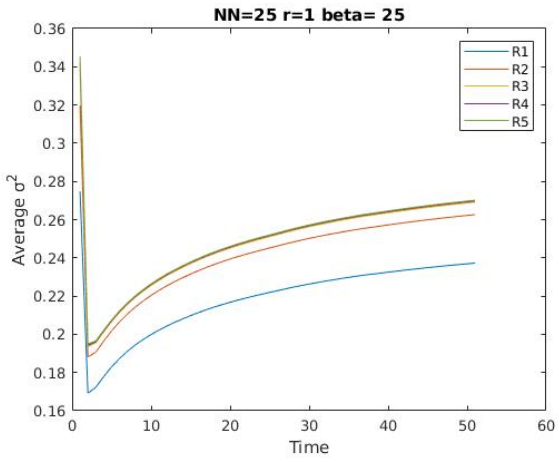
4.3 Concluding Remarks

By being a competent generative model for cell decision making in various biological contexts, the Least microEnvironmental Uncertainty Principle has shown good promise, in our continuing search for unifying principles in biological phenomena.

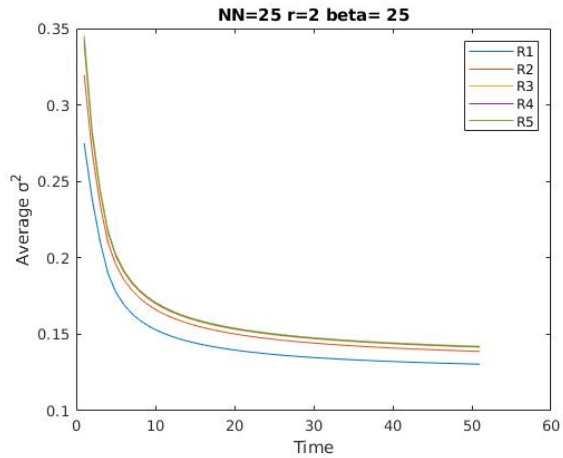
* * *

Chapter 5

Appendix

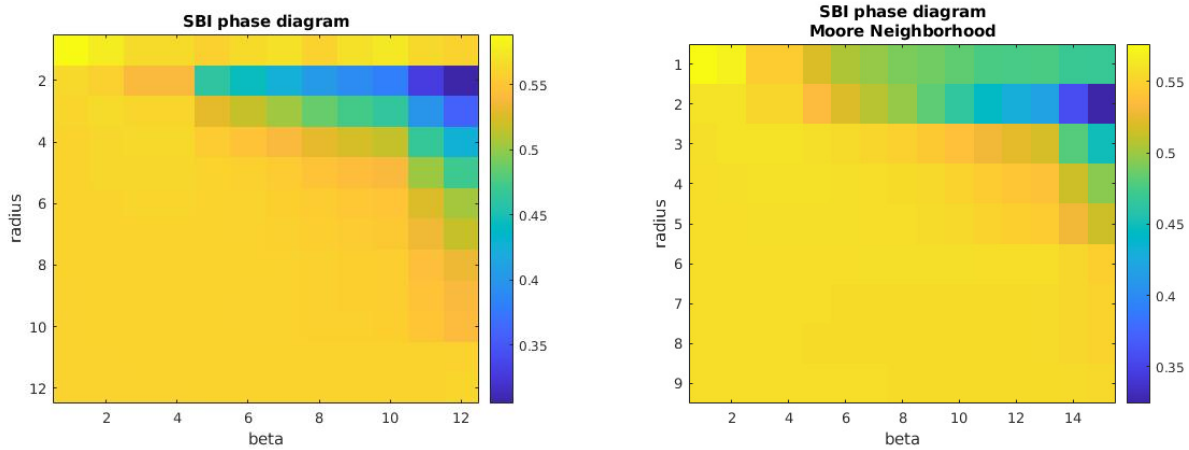


(a) Square Lattice, $(r = 1, \beta = 25)$



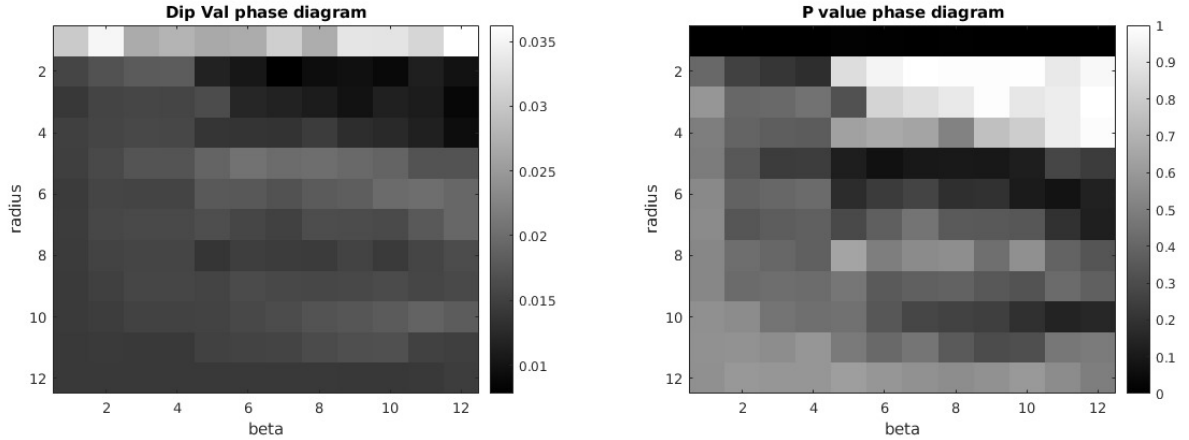
(b) Square Lattice, $(r = 1, \beta = 25)$

Figure 5.1: Saturating of $\langle \sigma^2 \rangle$ for $R = 3$ sized neighborhood



(a) Square Lattice, $\beta > 0$ (b) Hexagonal Lattice, $\beta > 0$

Figure 5.2: Phase Diagrams with Sarla Bimodality Index (SBI) as phase diagram. $SBI > 0.555$ meant multimodality, $SBI = 0.555$ meant uniformly random and $SBI < 0.555$ is for unimodal populations.



(a) Square Lattice, $\beta > 0$ (b) Square Lattice, $\beta > 0$

Figure 5.3: Phase Diagrams with Hartigan's Dip statistic and it's pvalue test (with null hypothesis as unimodal) as order parameters: larger the dip value, more bimodal the population. Larger the pvalue, more unimodal the population.

* * *

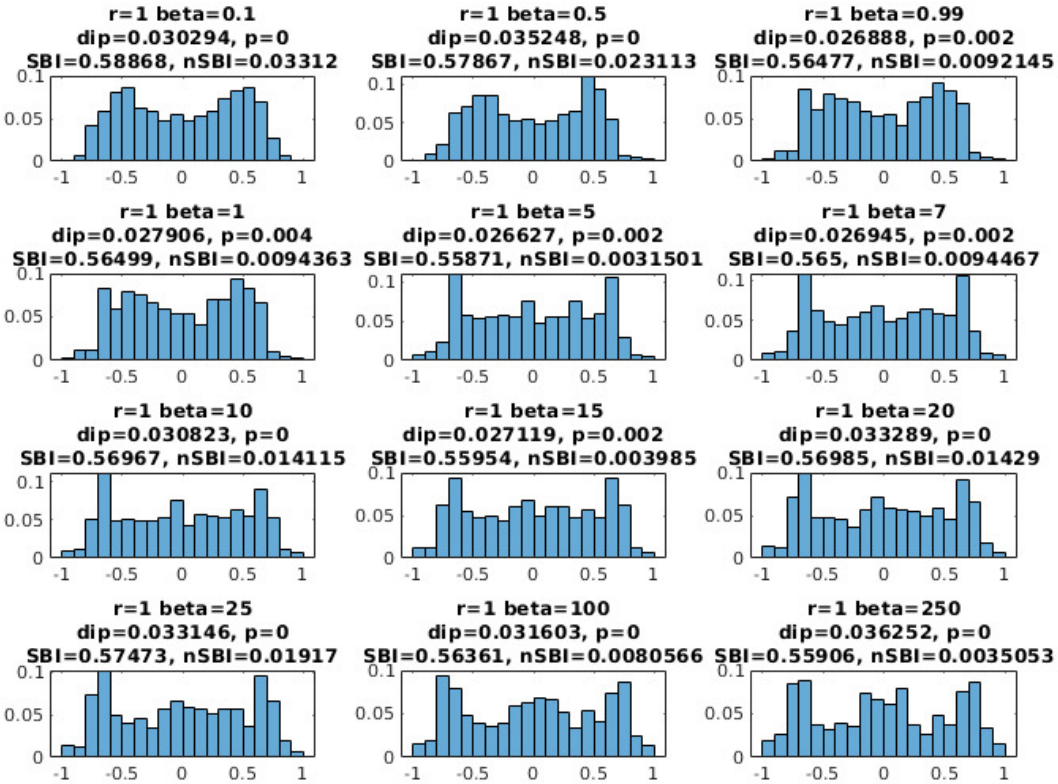


Figure 5.4: Square Lattice: Showing histograms and the index values of distributions ($PT2$,$PT3$) seen.

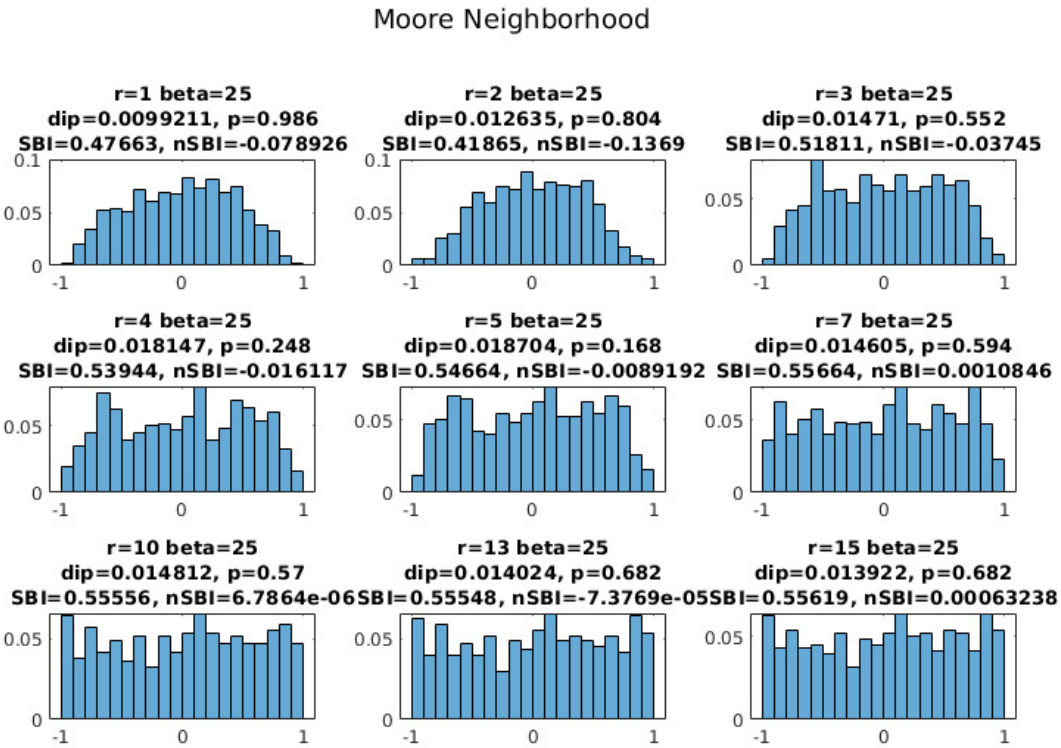


Figure 5.5: Hexagonal Lattice: Showing histograms and the index values of distributions ($PT2$) seen.

Bibliography

[1] Arnab Barua, Alireza Beygi, and Haralampos Hatzikirou. Close to optimal cell sensing ensures the robustness of tissue differentiation process: the avian photoreceptor mosaic case. *bioRxiv*, 2021.

[2] Arnab Barua, Josue M. Nava-Sedeño, Michael Meyer-Hermann, and Haralampos Hatzikirou. A least microenvironmental uncertainty principle (leup) as a generative model of collective cell migration mechanisms. *Scientific Reports*, 10(1):22371, Dec 2020.

[3] Arnab Barua, Simon Syga, Pietro Mascheroni, Nikos Kavallaris, Michael Meyer-Hermann, Andreas Deutsch, and Haralampos Hatzikirou. Entropy-driven cell decision-making predicts 'fluid-to-solid' transition in multicellular systems. *New Journal of Physics*, 22(12):123034, dec 2020.

[4] Marcelo Boareto, Mohit Kumar Jolly, Mingyang Lu, José N. Onuchic, Cecilia Clementi, and Eshel Ben-Jacob. Jagged–delta asymmetry in notch signaling can give rise to a sender/receiver hybrid phenotype. *Proceedings of the National Academy of Sciences*, 112(5):E402–E409, 2015.

[5] Federico Bocci, Jos Onuchic, and Mohit Kumar Jolly. Understanding the principles of pattern formation driven by notch signaling by integrating experiments and theoretical models. *Frontiers in Physiology*, 11, 07 2020.